

Enhancement of Performance of an EEG-based Brain-Computer Interface by means of a Time-Frequency Approach

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Abstract – We have developed a novel Time-Frequency approach of classification of motor imagery (MI) tasks for brain-computer interface (BCI) applications. Through off-line data analysis on data collected during a “cursor control” experiment, we evaluated the capability of our proposed method in revealing the major features of the EEG control and enhancing MI classification accuracy. The pilot results in two human subjects are promising, with a mean accuracy rate of 87.9%, suggesting the feasibility of defining a new and more reliable EEG-based BCI.

I. INTRODUCTION

The ultimate goal of brain-computer interface (BCI) techniques shall be to provide those people with severe motor disabilities alternative means of communication and control. Typically, a BCI system extracts, from scalp-recorded EEG, particular components or features encoding human intention and conveys the resulting control signals to the external world. One type of brain-computer interfaces is based on the detection and classification of the change of EEG rhythms during different motor imagery (MI) tasks, such as imagination of left- and right-hand movements. The accuracy of MI classification directly determines the performance and reliability of such BCI applications, and thus is of great importance. Recently, we have developed a new approach in our laboratory for motor imagery classification by means of a weighted time-frequency scheme in an attempt to define a novel and improved EEG-based BCI [1-2].

In the present study, the off-line analysis was conducted on data collected during performance of “cursor control” by two subjects, and a time-frequency synthesized spatial pattern was revealed on each individual subject during MI tasks. Furthermore, we proposed a time-frequency approach and evaluated its capability in enhancing the accuracy of motor imagery classification in comparison with the online performance using BCI2000 [3].

II. METHODS

A. Data description

The scalp EEG signals were recorded from 32 electrodes, placed over the upper hemisphere with a sampling rate of 200Hz, during an online cursor control experiment utilizing the BCI2000 – a general-purpose system for BCI research [3]. Specifically, a subject sat at a monitor and watched a cursor move continuously from left to right across the screen. A vertical target bar randomly appeared near either the upper-right corner or the lower-right corner. The subject attempted to deflect the cursor up or down to hit the bar by imagining left- or right-hand movement, which was accomplished by an online BCI

process that responded to subject’s intention based on mu rhythm (8-12Hz) features [3,4]. The off-line data analysis on two subjects (one male and one female, both aged at 23 years) was presented in this paper. Nine electrodes over the somato-sensory cortex (the primary cortical region related with motor-imagery neural activity) were selected from the 32 electrodes for off-line data analysis.

Each trial, during the time course of cursor moving from left to right, started with a 1 second period in which the target appeared on the screen without the cursor. During the subsequent 4 seconds, the cursor appeared and moved across the screen, with the subject performing the left- or right-hand motor imagery task. The next 1 second time interval consisted of both the cursor and the target remaining on the screen if the subject successfully ‘hit’ the target, else in the case of a ‘miss’, the cursor remained and the target disappeared. The final 1 second interval consisted of a blank screen.

The subjects completed 480 trials evenly divided between left- and right-hand imagination. The experiment was carried out twice a week on each subject for a minimum of three weeks.

B. Preprocessing

During the off-line data analysis, the recorded EEG signals were sequentially preprocessed by applying surface Laplacian filtering, frequency decomposition and ERD/ERS feature extraction.

Scalp recorded EEG represents the noisy spatial overlap of activities arising from very diverse brain regions. Surface Laplacian filtering attempts to accentuate localized activity and reduce diffusion in multi-channel EEG. Assuming that the distances from a given electrode to its four directional neighboring electrodes are approximately equal, the surface Laplacian can be approximated by subtracting the average value of the neighboring channels from the channel of interest as Eq(1).

$$V_j^{Lap} = V_j - \frac{1}{n} \sum_{k \in S_j} V_k \quad (1)$$

where V_j is the scalp potential EEG of the j th channel, and S_j is an index set of the four neighboring channels.

The EEG component from 6 to 30 Hz was further decomposed into multiple frequency bands using a constant-Q (also called proportional-bandwidth) scheme [1]. Specifically, we constructed a set of fourth order Butterworth band-pass filters, each of which span the indicated octaves with the ratio (Q) of the center frequency to the bandwidth being chosen to be a constant. The neighboring frequency bands had certain overlapping to

allow a proper redundancy of signals.

We delineated the event-related desynchronization (ERD) and the event-related synchronization (ERS) features [5] within each frequency band by extracting the envelop of instantaneous powers of decomposed signals [1]. The ERD/ERS features were further simplified by down-sampling the envelop, since the high frequency (>5Hz) component can be ignored for most envelops (Details see [1]).

C. Classification by correlation of spatial patterns [1]

A vector $p(t, f)$ used in the classification is composed of 9 potential values of 9 electrodes. A value of an element of the vector is the potential value of an envelope at a time and a frequency for an electrode. Two types of averaging vectors, i.e. $P_L(t, f)$ and $P_R(t, f)$, were made using training data to calculate the correlation, where the subscript L or R stands for the tasks of left or right hand imagined movements, and t and f stand for time and frequency respectively. Correlation coefficients $C(t, f) = C(p(t, f), P(t, f))$ were calculated using (2).

$$C = \frac{(p - \bar{p})^T (P - \bar{P})}{\|p - \bar{p}\| \cdot \|P - \bar{P}\|} \quad (2)$$

Where $\bar{p}$ and $\bar{P}$ denote the mean values of p and P respectively. The classification of left or right was determined by the value derived from (3), i.e. $g(p)=1$ and -1 denote left and right respectively.

$$g(p) = \text{sgn} \left[\sum_{t=1}^T \sum_{f=1}^F w_{t,f} h_{t,f}(p) \right] \quad (3)$$

Where, T and F denote the sampling numbers in time and frequency domains, and

$$h_{t,f}(p) = \text{sgn}[\delta_{t,f}], \quad (4)$$

$$\delta_{t,f} = C(p, P_L) - C(p, P_R). \quad (5)$$

$w_{t,f}$ are the time-frequency weights [1] defined to emphasize $h_{t,f}(p)$ that are expected to contribute to good classification accuracies. However, as described in [1], use of $w_{t,f}$ decreased the accuracy rates in this study as well. Therefore, $w_{t,f}$ was not used actually, i.e. $w_{t,f} = 1$ for all $h_{t,f}(p)$.

D. Use of summation of $\delta_{t,f}$

We changed the equation from (3) to (6) in order to examine the difference of in the accuracy rates between (1) and (6).

$$g(p) = \text{sgn} \left[\sum_{t=1}^T \sum_{f=1}^F \delta_{t,f} \right] \quad (6)$$

E. Classification by correlation of comprehensive vectors with time-frequency spatial information

We introduced a comprehensive vector with time-frequency spatial information. The algorithm of this method is as follows. First, the same P_L and P_R vectors in section C are made, and the values of the correlation coefficients between two envelopes with the same frequency consisting of P_L and P_R are calculated. When the values are larger than a threshold C_0 , the corresponding envelope data are deleted from both P_L and P_R . The value of C_0 is determined to give a good classification accuracy in the training data. Next, the remained envelope vectors are connected to be a row vector for P_L and P_R , and the two reconstructed vectors are denoted $P_{c,L}$ and $P_{c,R}$ respectively. The test vectors, p_c , are constructed using its envelope data that are the same frequency and electrode as P_L and P_R . The classification of left or right is determined by the value derived from (7), i.e. $d=1$ and -1 denote left and right respectively.

$$d = \text{sgn}[C(p_c, P_{c,L}) - C(p_c, P_{c,R})] \quad (7)$$

III. RESULTS

The classification accuracy rates are calculated using (3), (6) and (7) for 5sec. The number of sessions used is 29 for subject-1 (S1) and 30 for subject-2 (S2). The number of training sessions is 14 for S1 and 15 for S2, and the others are used for test data respectively. The results are shown in Table 1. The values in the column denoted (7)-1 are the accuracy rates for the case without deleting envelopes as described in section E, this case is the same as that of $C_0 = 1$. The values in the parenthesis of column (7)-2 are the value of threshold C_0 .

TABLE I
ACCURACY RATES [%] OF 2 SUBJECTS

	BCI2000	(3)	(6)	(7)-1	(7)-2
S1	83.3	85.0	88.8	85.4	90.4 (0.83)
S2	82.9	85.5	82.1	85.4	85.4 (0.98)
Mean	83.1	85.3	85.5	85.4	87.9

The rates calculated by (6) and (7) are compared with that of BCI2000 and (1). The method using (6) enhanced the rate of S1 and conversely worsened that of S2. The method of section E using (7) and the threshold ((7)-2) enhanced the rate of S1 but did not dramatically enhance that of S2. The rate of S2 was almost the same value, about 85%, as that of (1). Compared to the original (BCI2000) result, the present method using (7) enhanced the mean accuracy rate of the two subjects for the tasks of left or right hand imagined movements.

IV. DISCUSSION

The present study suggests that by using the time-frequency approach we can further enhance the performance of an EEG-based Brain-Computer Interface. When the proper threshold value is determined, this method has the possibility to return a good accuracy rate. Our pilot study in two human subjects indicates an increase in the classification rate from 83.1% to 87.9%. An additional advantage of this method is that the calculation time is short, suitable for implementation in on-line experiments. In summary, this pilot study is promising and suggests the time-frequency approach merits further investigation to enhance the classification accuracy of EEG-based BCI systems.

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